Avishkar AV

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Professional Summary

Fullstack Software Engineer with production experience architecting high-concurrency microservices, geospatial data engines, and scalable cloud platforms. Sole backend architect of bluedot.ngo (FastAPI, PostGIS, Cashfree 80G tax compliance) and core engineer of Passiton serving 13,000+ active users on GCP Cloud Run. Strong background in database query tuning, sub-50ms Redis caching, Docker CI/CD pipelines, and regulatory data compliance (DPDP/GDPR).

Work Experience

Fullstack Developer, Third Eye Creative | Dec 2025 – Present | Bengaluru, India

- Passiton (13,000+ Active Users): Shipped and maintained the peer-to-peer recycling marketplace on GCP Cloud Run with autoscaling; architected dual ASGI server running FastAPI for mobile and Flask for admin.
- BlueDot (Solo Backend Architect): Architected the entire backend for bluedot.ngo as an asynchronous FastAPI REST API backed by PostgreSQL (Supabase) with SQLAlchemy 2.0 ORM and Alembic migrations.
- Database Performance Tuning: Shrank production PostgreSQL database footprint 60% (2GB to 800MB) and accelerated complex join queries 40% through zero-downtime poison-pill row cleanup and targeted indexing.
- Latency & Cost Optimization: Slashed feed response latency from 10s to under 2s with a Redis-cached feed algorithm, and eliminated 100% of third-party map costs by integrating OpenStreetMap.
- PostGIS Geospatial Engine: Stored restoration site boundary polygons (SRID 4326) and built automated PostgreSQL database triggers with ST_Contains to increment cached tree counts without runtime aggregates.
- FinTech & 80G Tax Compliance: Integrated Cashfree Payment Gateway supporting webhook signature verification, delayed PAN collection UX flow, and automated 80G tax receipt PDF generation.
- AI Reliability Engineering: Cut Vertex AI API failure rate from 33% to 3% with exponential backoff retry logic; integrated dual-engine AI vision (Vertex AI & PI@ntNet) with 70% threshold auto-publishing.
- Cloud Infrastructure & CI/CD: Cut infrastructure spend 46% by migrating legacy cash-collection tool from AWS EC2 to DigitalOcean droplets with Docker; built multi-tenant CI/CD with GitHub Actions.

Fullstack Developer Intern, EarthlyExotica | Jan 2025 – May 2025 | Bengaluru, India

- Built mobile-responsive frontend in Flutter for baadige.com utilizing modular UI components and integrated core REST APIs connecting platform features.

Technical Skills

Languages & Backend: Python, SQL, JavaScript, TypeScript, Dart, Java, FastAPI, Flask, SQLAlchemy, Alembic, Pydantic, REST APIs, JWT, RBAC

Databases & Cloud: PostgreSQL, PostGIS (ST_Contains, ST_DWithin), Redis, Supabase, MySQL, MongoDB, GCP (Cloud Run, Vertex AI), AWS (EC2), Docker, CI/CD

FinTech & Tools: Cashfree Payment Gateway (Webhooks, 80G Receipts), MSG91, OpenStreetMap, Git, Linux, Postman

Technical Projects

- Autonomous YouTube Video Publishing Engine: Engineered a zero-cost serverless media pipeline publishing daily 1080p Shorts to YouTube via GitHub Actions CI/CD; automated OAuth 2.0 refresh-token handshake and resumable chunked video uploads (PUT protocol) using 16% of daily free quota, paired with Gemini Flash AI scriptwriting, local Kokoro-82M ONNX TTS, Faster-Whisper alignment, and 14 Mbps headless Chrome CDP WebGL rendering (Python, Node.js, FFmpeg, YouTube Data API v3, Supabase).
- Smart Attendance System: Real-time facial recognition pipeline on CCTV feeds replacing manual logging with automated CSV export (Python, OpenCV, NumPy).
- Weather Alert & SMS Notification System: Automated daemon broadcasting localized rainfall alerts via SMS from real-time meteorological sensor feeds (Python, Twilio API).

Education & Certifications

B.E. in Computer Science & Engineering, Visvesvaraya Technological University (VTU) | Dec 2021 – June 2025 | CGPA: 7.95 / 10.0

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional • Oracle OCI AI Foundations Associate • Java OOP Certification